

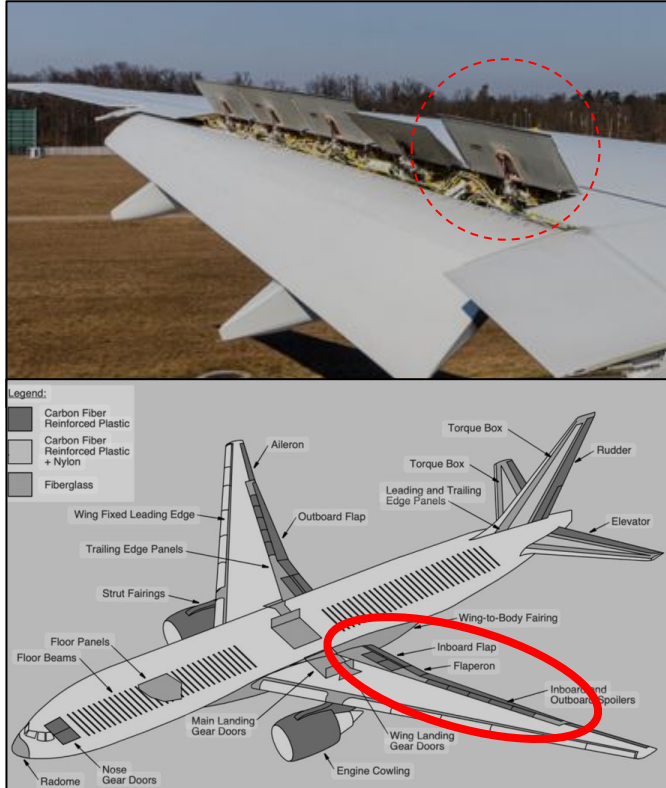
# Composite Wing Spoiler Component Non-compliances

Group 11: Jean Kim, Aryan Partabian, Anjali Roy, Simon Grant, Daniel Jung

*Mentors: Joe Pruss, Rob Albers*



# Project Context



## Wing Spoiler Quality Challenge

### Our Scenario

- Product: Wing spoilers (CFRP construction)
- Critical aerodynamic surface (Paint Finish) on upper wing

**The Problem:** Quality Assurance rejected spoilers for:

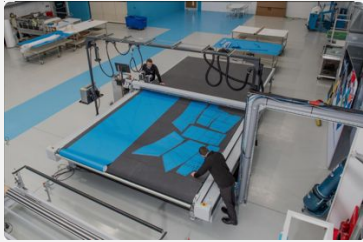
- Tool side surface depressions
- Core edge depressions
- Finish adhesion failures
- Finish thickness variations

**Business Impact:** Assess the long-term impact and suggest modifications to the manufacturing process.

# Composite Manufacturing Process Overview

1

## Material Preparation



- Prepreg storage & kitting
- Temperature-controlled cutting

2

## Tool Setup



- Cleaning & release agent
- CTE-matched tools

3

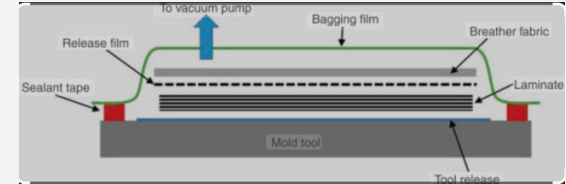
## Lamination



- Manual or automated lay-up
- Ply-by-ply stacking

4

## Cure



- Vacuum Bagging
- Autoclave cycle
- Controlled temperature/pressure

5

## Post-Processing

- NDI (ultrasound, X-ray)
- Trimming & drilling
- Assembly

6

## Finishing

- Surface preparation
- Primer & topcoat



# Tool Side Depression Fishbone

## Cure / Post-Cure

Autoclave

Temperature / Pressure

Part handling beyond cure

## Tooling

Tooling mold surface defects

Tool dimensional deviation

Tool history

Cleaning / prep material storage

Release agent storage/age

Prepreg management

## Material Handling

Release application

Prepreg lay up

Mold cleaning procedure

## Staff Training

## Root Cause Identification:

- Compare Available Records  
(Tool use, maintenance, inspections)
- Staff methods of mold cleaning and release agent removal

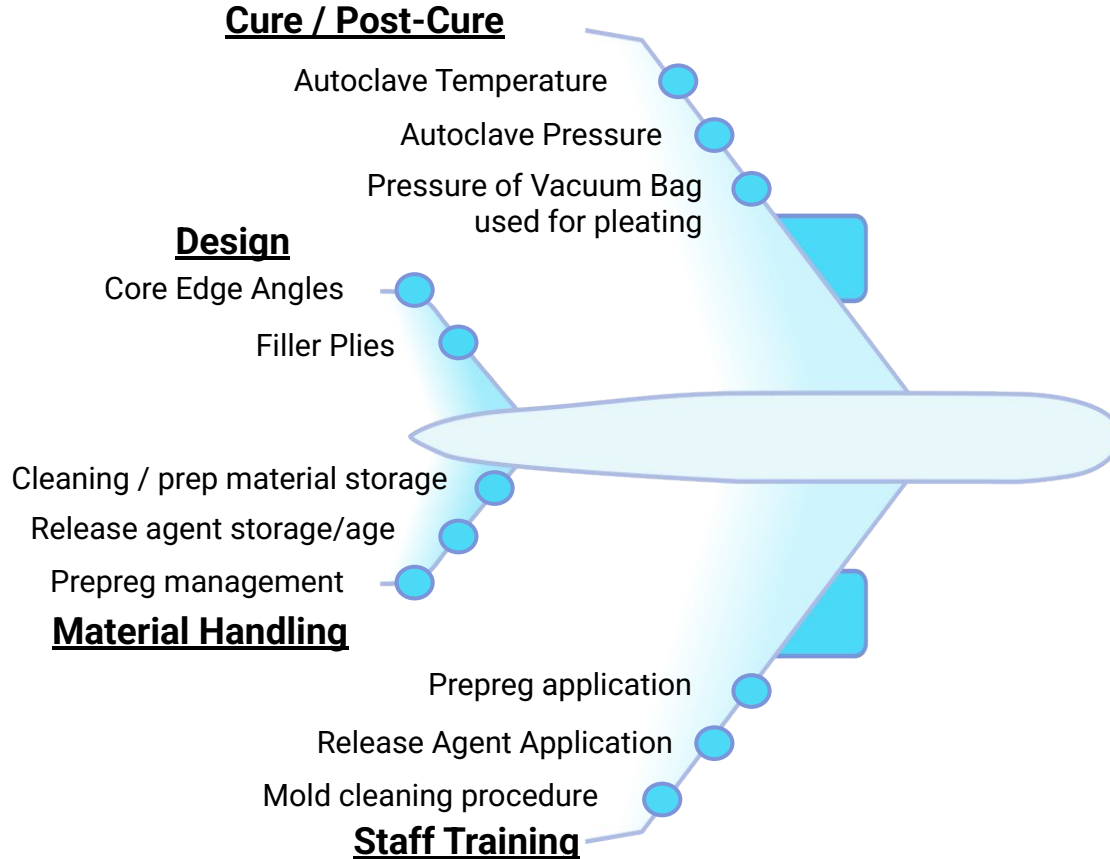
## Results:

- Tools are within spec
- Release agent/prepreg applied properly
- Release agent on mold is not stripped properly

# Tool Side Depression Analysis & Solution Diagram

|                             | Economic Concern |            |            | Sustainability | Flow time |
|-----------------------------|------------------|------------|------------|----------------|-----------|
|                             | Personnel        | Equipment  | Process    |                |           |
| Tool Preparation            | \$               |            | \$\$       | ⚠️⚠️           | 🕒         |
| Establish a Stripping Cycle | \$               | \$\$       | \$\$       | ⚠️             | 🕒🕒        |
| Tooling Maintenance         | \$               |            | \$         |                | 🕒         |
| Tool Preparation Automation | \$\$\$           | \$\$\$\$\$ | \$\$\$\$\$ | ⚠️             | 🕒🕒🕒🕒      |

# Core Edge Depression Fishbone



## Root Cause Identification:

- Analysis of existing records (material handling, cure process)
- Testing of different vacuum bag pressures
- Prototype design changes

## Results:

- Pleating of the vacuum bag is the most significant contribution to the depressions

# Core Edge Depression Analysis & Solution Diagram

|                                        | Economic Concern |           |          | Sustainability                                                                      | Flow time                                                                           |
|----------------------------------------|------------------|-----------|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                        | Personnel        | Equipment | Process  |                                                                                     |                                                                                     |
| Staff Training<br>(Pleating Technique) | \$               |           | \$       |  |  |
| Design Change<br>(Core Angle)          | \$\$\$\$         | \$        | \$\$\$\$ |                                                                                     |  |
| Design Change<br>(Filler Ply)          | \$\$\$\$         |           | \$\$\$\$ |                                                                                     |  |

# Paint Adhesion Fishbone

## Surface Contamination

Silicone residues

Fingerprints

Mold release residues

## Staff Training

Improper sanding technique

Improper use of solvent wipe

Incompatible primer with substrate

## Chemistry

Improper grit for sanding

Faulty solvent wipes

## Equipment

## Root Cause Identification:

- Compare Available Records (Environmental, Personnel, and equipment) to adherence results (ASTM D3359)
- Surface test to identify possible contaminants (FTIR)

## Results:

- Adhesion issues caused by improper surface preparation
- Improper grit used (400)

# Paint Adhesion Analysis & Solution Diagram

|                                  | Economic Concern |           |          | Sustainability | Flow time |
|----------------------------------|------------------|-----------|----------|----------------|-----------|
|                                  | Personnel        | Equipment | Process  |                |           |
| Surface Prep Training            | \$\$             |           | \$\$     | ⚠️⚠️           | 🕒🕒        |
| Establishing visual standards    |                  | \$        | \$       |                | 🕒         |
| Change grit size to 180 from 400 |                  | \$        | \$       |                | 🕒         |
| Automating Process               | \$\$\$           | \$\$\$    | \$\$\$\$ | ⚠️             | 🕒🕒🕒🕒      |

# Paint Thickness Fishbone

## Environment

- Temperature could be too hot/cold or humid/dry
- Part could be contaminated
- Might be exposed to UV

## Staff Training

- Staff could make passes too slow
- Staff may not know right thickness

- New paint could be more viscous
- Paint may be expired

## Chemistry

- Pressure may be wrong
- Uneven flow rate
- Nozzle could be wrong size

## Equipment

## Root Cause Identification:

- Investigate environment
- Investigate paint composition
- Check what equipment is being used
- Watch staff to see where problems arise

## Results:

- Environment wasn't the cause
- Paint was good & unchanged
- Inconsistencies with nozzle and staff technique

# Paint Thickness Analysis & Solution Diagram

|                         | Economic Concern |           |         | Sustainability                                                                      | Flow time                                                                           |
|-------------------------|------------------|-----------|---------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                         | Personnel        | Equipment | Process |                                                                                     |                                                                                     |
| Retrain Staff           | \$               |           | \$\$    |  |  |
| Standardize Nozzle Size |                  | \$        | \$      |                                                                                     |  |
| Automating Process      | \$\$             | \$\$\$\$  | \$\$\$  |  |  |

# Summary of Solutions

|                                | Non Conformances             |                                |                                                    |                             |
|--------------------------------|------------------------------|--------------------------------|----------------------------------------------------|-----------------------------|
|                                | <u>Paint Adhesion</u>        | <u>Paint Thickness</u>         | <u>Tool Side Depression</u>                        | <u>Core Edge Depression</u> |
| <b>Staff Training</b>          | Surface Preparation Training | Application technique training | Tool preparation training                          | Pleating technique training |
| <b>Process Standardization</b> | Visual standards             | Nozzle size                    | Mold stripping cycle & Regular tooling maintenance | -                           |

**Any questions?**

# Sources

Al-Fatlawi, Alaa, et al. "Optimal Design of a Fiber-Reinforced Plastic Composite Sandwich Structure for the Base Plate of Aircraft Pallets In Order to Reduce Weight." MDPI, Multidisciplinary Digital Publishing Institute, 9 Mar. 2021

Black, Sara. "Getting To Know ". Mmsonline, 8 Oct. 2008, <https://mmsonline.com/articles/getting-to-know-black-aluminum>.

"Boeing 737 Production". B737

Boeing Commercial Airplane Group, and NASA Langley Research Center. *Boeing/NASA Composite Components Flight Service Evaluation: Final Progress Report*. NASA, 1989. NASA Technical Reports Server, [ntrs.nasa.gov/api/citations/19900013293/downloads/19900013293.pdf](https://ntrs.nasa.gov/api/citations/19900013293/downloads/19900013293.pdf).

Gardiner, Ginger. "3D-printed CFRP tools for serial production of composite landing flaps - CEAD | Large Scale Additive Manufacturing". Ceadgroup.

Meindlhumer, Martin, et al. *Journal of Aircraft*, vol. 56, no. 1, Jan. 2019, pp. 85–93, <https://doi.org/10.2514/1.c034891>.

"Our Facilities - Composite Manufacturing for OEM". Pirancomposites.

"Specialist surface finishing systems". Expoworldwide, <https://expoworldwide.com/case-studies/specialist-surface-finishing-systems>.

"Spoilers." IVAO Documentation Library, [wiki.iviao.aero/en/home/training/documentation/Spoilers](https://wiki.iviao.aero/en/home/training/documentation/Spoilers). Accessed 31 Oct. 2025.

Xiaomei, Fang. "Effectiveness of Surface Treatment Techniques." NASA, NASA, [ntrs.nasa.gov/](https://ntrs.nasa.gov/). Accessed 5 Dec. 2025.